

		Hence, 4.7 cm is a logical choice.
6	B	Output power of wind turbine generator $P_o = \frac{1}{2} \frac{\Delta m}{\Delta t} (v_i^2 - v_f^2) \times \text{efficiency of generator}$ $= \frac{1}{2} (9.7) (4.0^2 - 1.5^2) (0.60)$ $= 40.01 \text{ W}$ Electrical power $P = IV$ Option A: $12 \times 1.5 = 18$ (not possible) Option B: $12 \times 3.3 = 39.6$ (possible) Option C: $24 \times 0.60 = 14.4$ (not possible) Option D: $24 \times 2.8 = 67.2$ (not possible)
7	C	It takes one year for the Earth to complete one orbit around the Sun, with